

Final Design Review

Team Rock-E

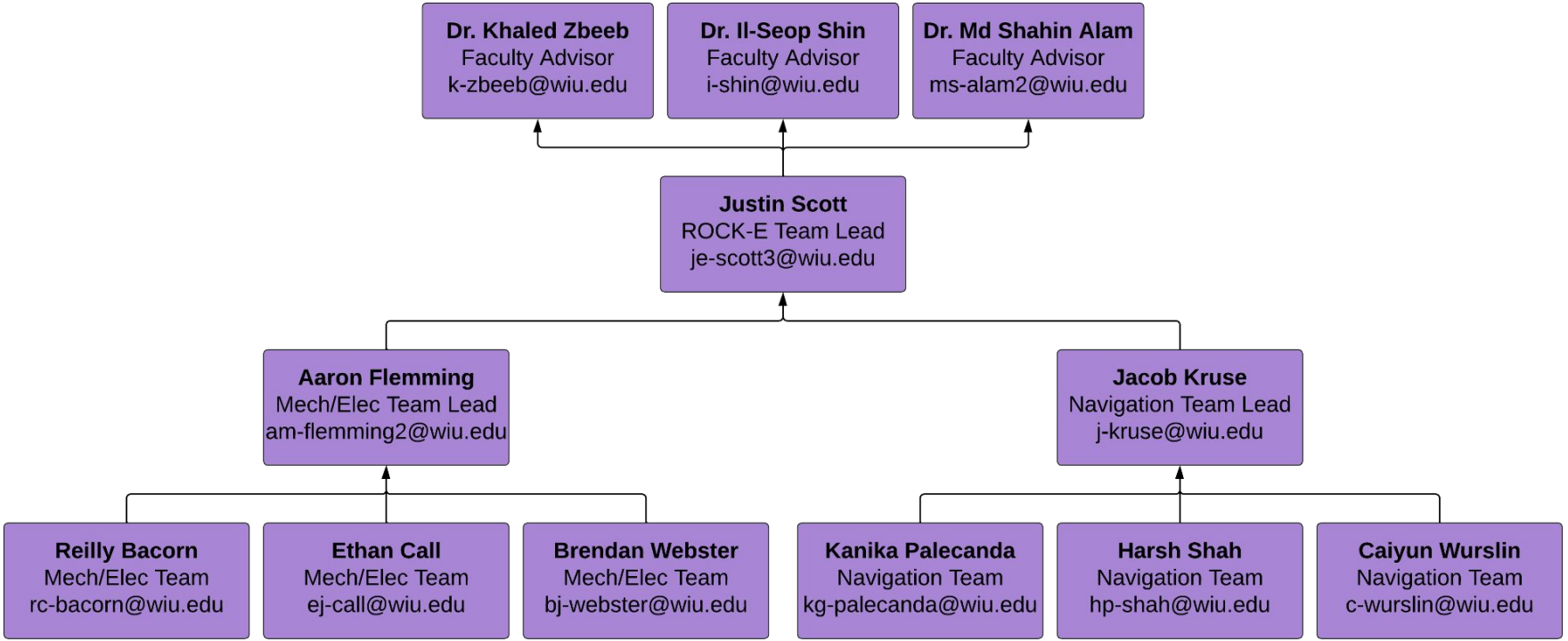
May 1, 2024

Rock-E Team Members

Harsh Shah	Mechanical Engineering
Brendan Webster	Mechanical Engineering
Reilly Bacorn	Mechanical Engineering / Engineering Physics
Kanika Palecanda	Mechanical Engineering
Justin Scott	Mechanical Engineering
Caiyun Wurslin	Electrical Engineering
Ethan Call	Mechanical Engineering
Jacob Kruse	Electrical Engineering
Aaron Flemming	Mechanical Engineering



Team Breakdown



Western Illinois University
SCHOOL OF ENGINEERING & TECHNOLOGY

What are we designing?

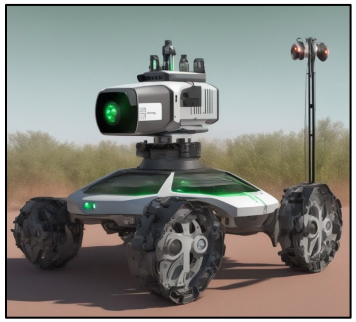
- Vehicle to Compete in 2024 IGVC
- Showcase Vehicle for WIU SET





Why an Autonomous Vehicle?

- Fully Adopted by 2030
- Skills Needed by Industry

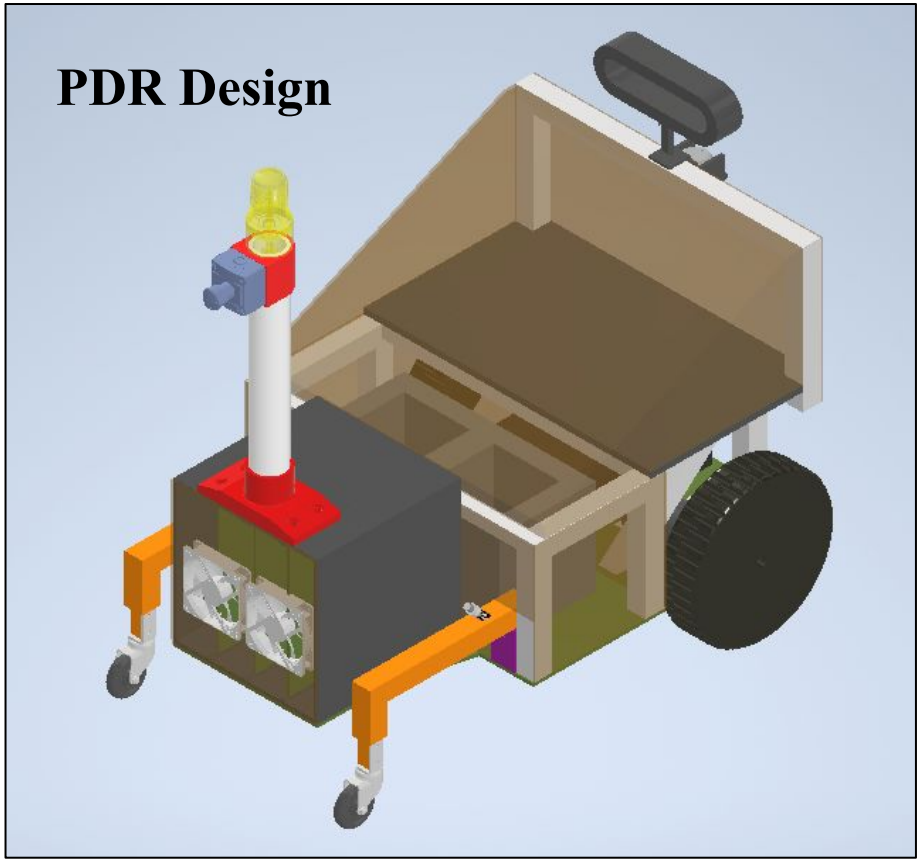


What are the parameters of our design?

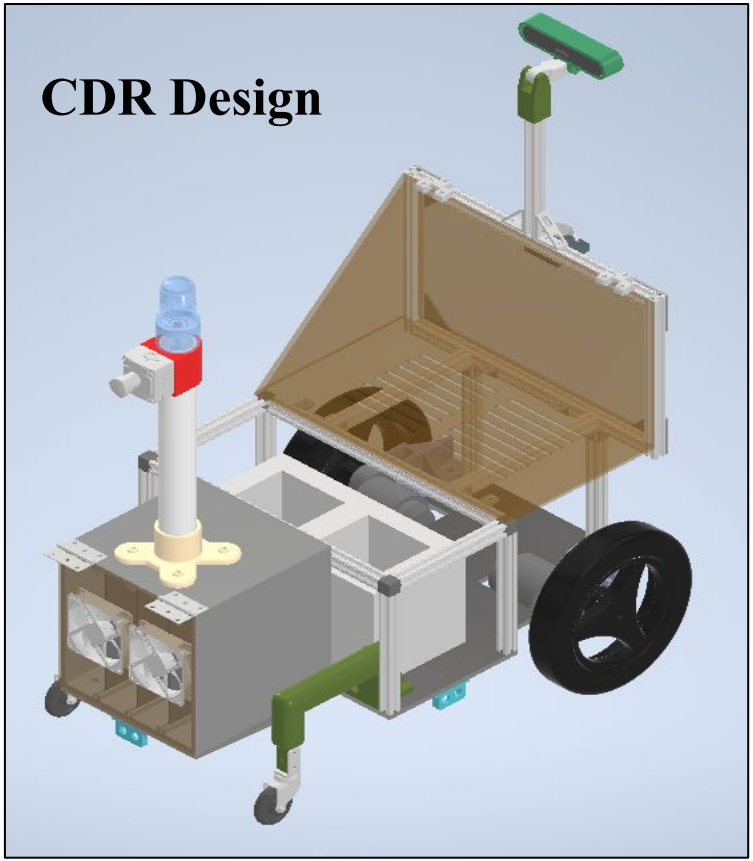
- 2024 IGVC rules - Appendix S
- Budget $5,000 + Fundraising
- May 2024

PDR/CDR CAD Designs

PDR Design



CDR Design



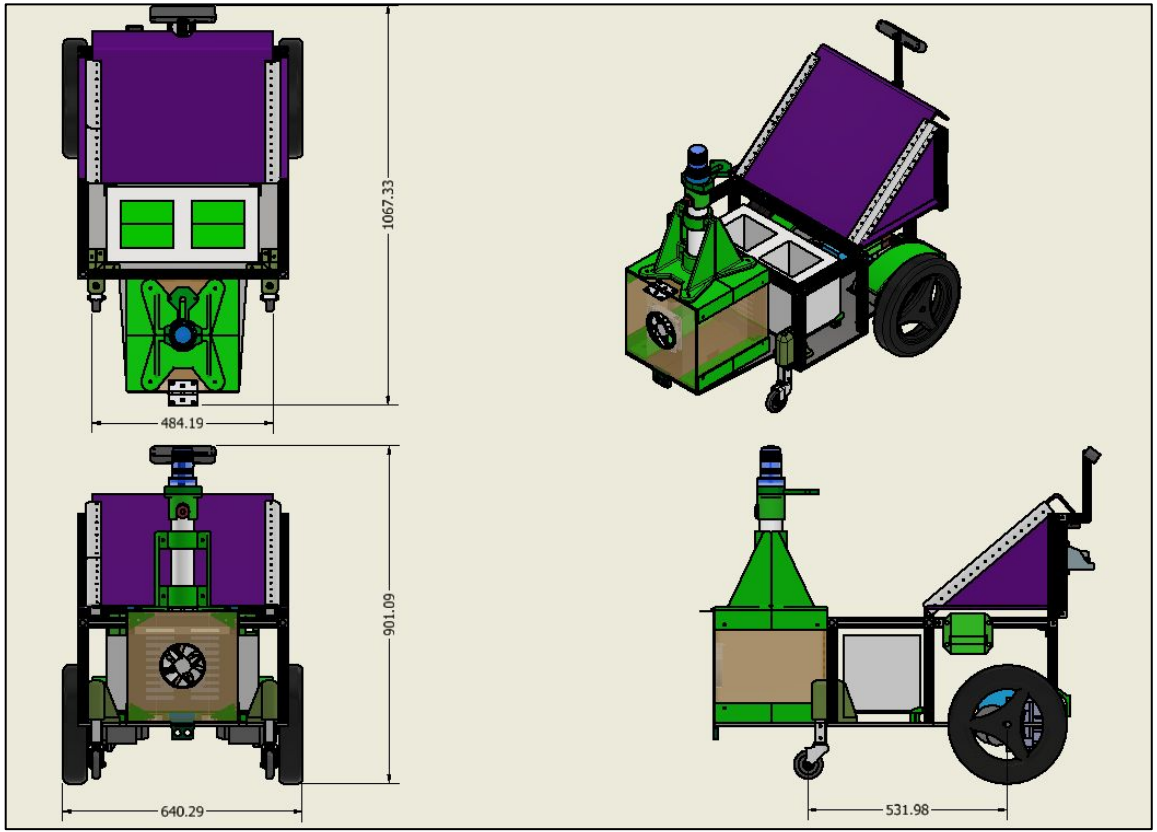
FDR CAD Design





Rock-E Dimensions

- **Length: 1.067m**
  (Required: 0.914m)
- **Max Width: 0.64m**
  (Required: 0.61m)
- **Max Height: .901m**
  (Required: Under 1.82m)
- **Width Between Casters: 0.484m**
- **Max Measured Speed: 3.6 mph**
  (Required: Under 5 mph)
- **Dry Weight: 39.8kg**
- **Gross Weight: 57kg**

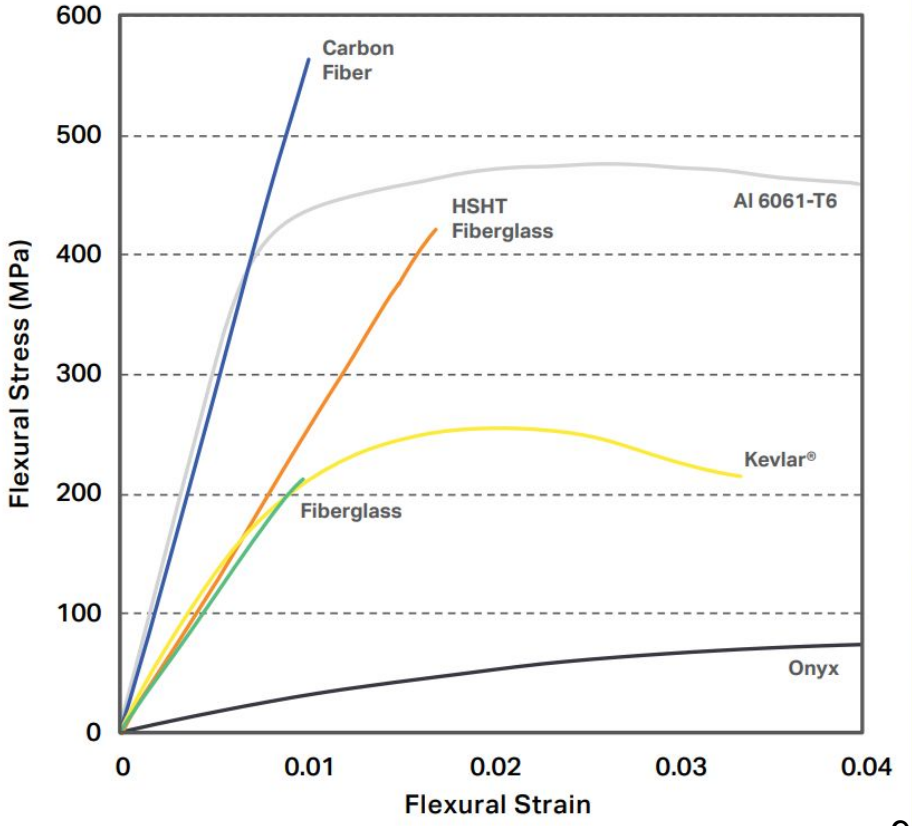


Exploded View



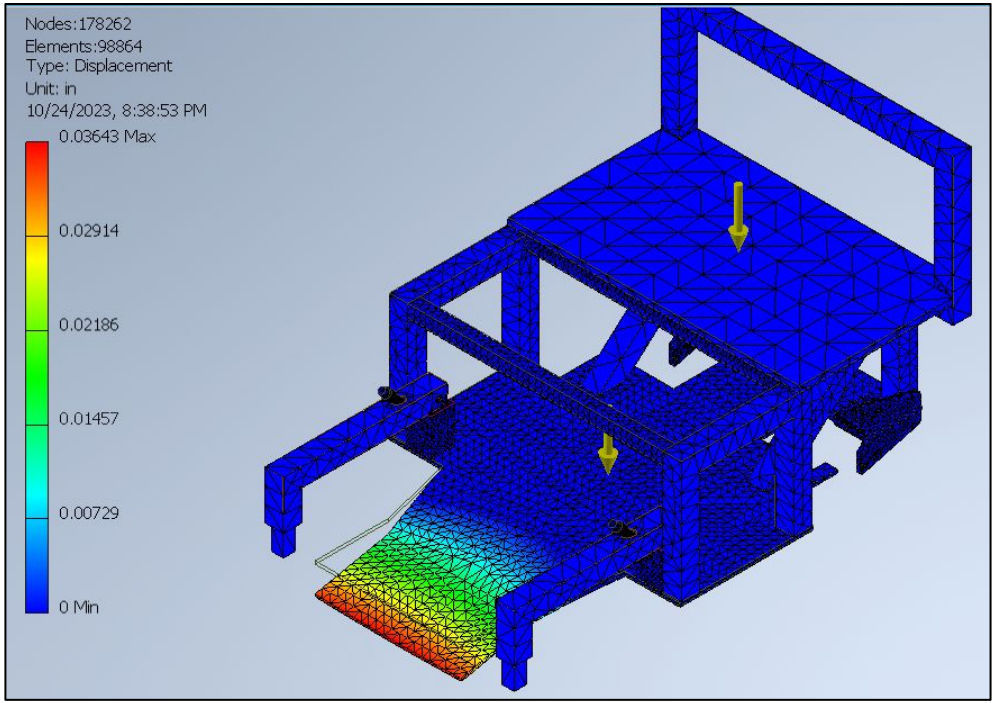
Material Choices

	Minimum yield strength MPa	Benefits
6061 Aluminum	240	Strong, lightweight, inexpensive, machineable
PLA	26	Inexpensive, additive
PETG	57	Flexible, additive
Onyx (with Carbon Fiber)	240	Strong, lightweight, additive
Acrylic	54	Transparent, machinable

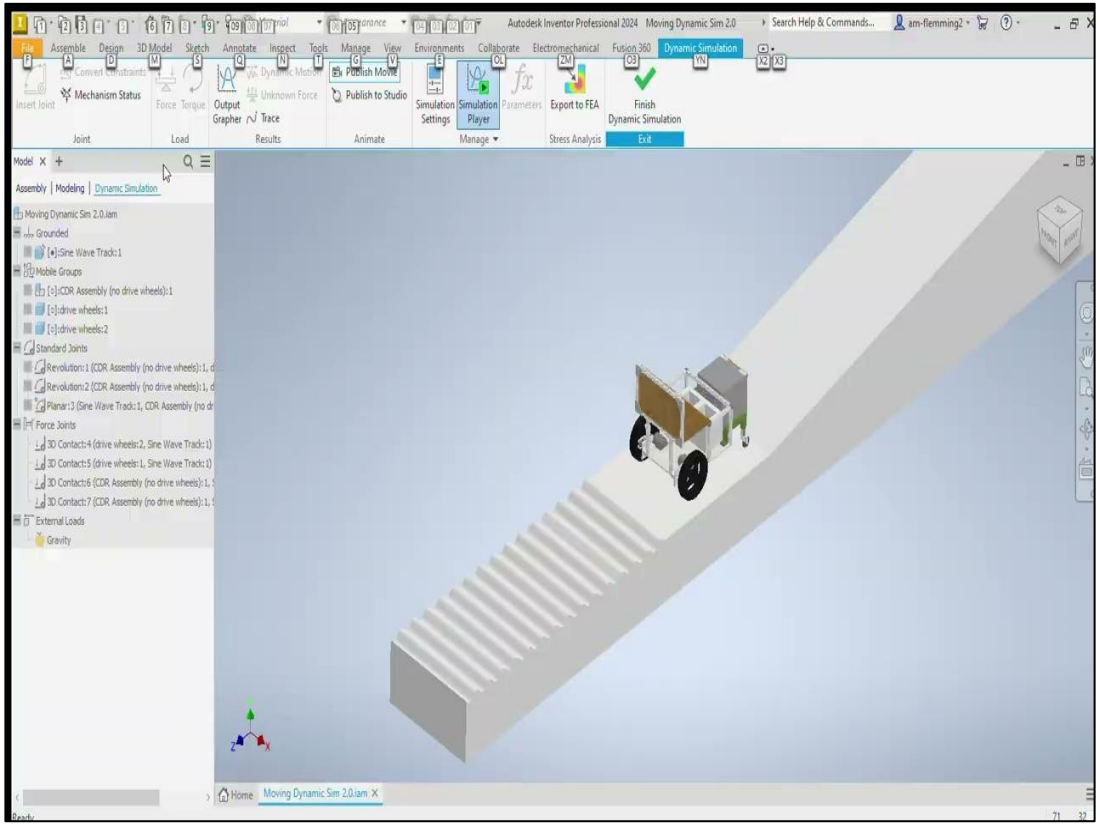


Static Structural Analysis

- Maximum displacement: **0.9253 mm**
- Maximum stress: **9.88 MPa**
- Base material rated for **241.32 MPa**
- Assumed **66.72 N** (15 lb) laptop (3.08 lb actual)
- Assumed **444.82 N** (100lb) force on base



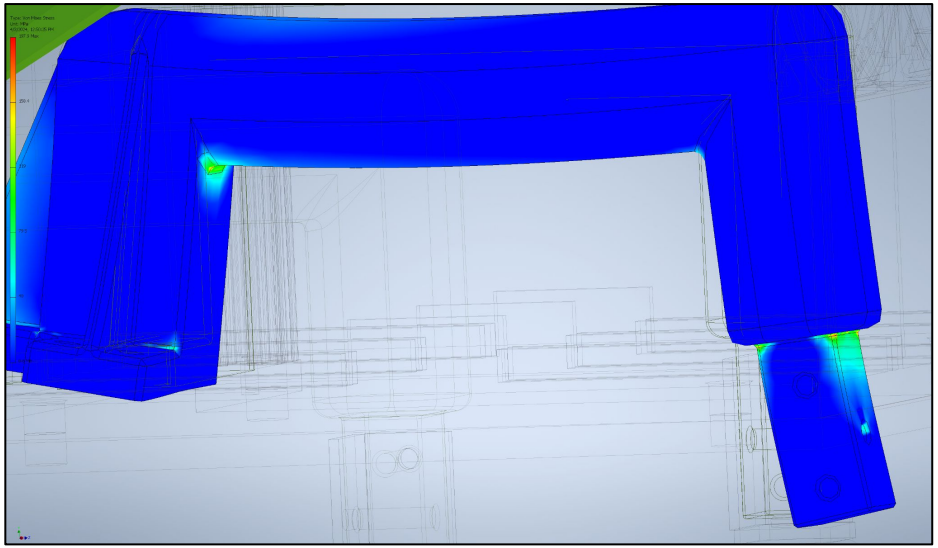
Dynamic Simulation Analysis



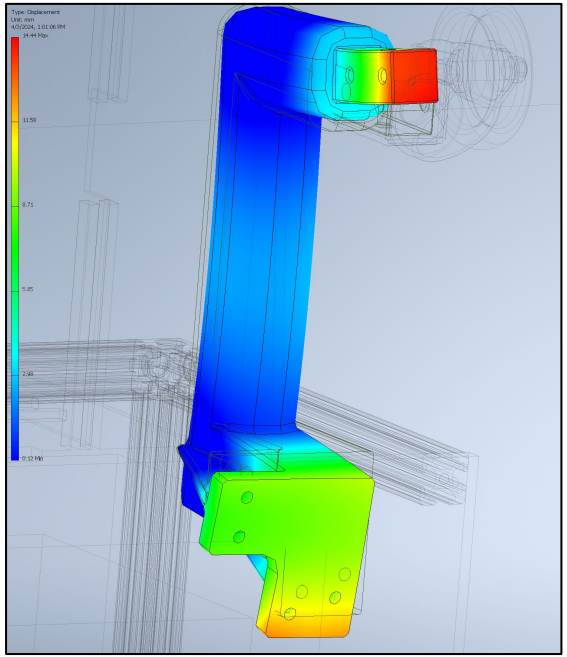
Dynamic Simulation Analysis: Suspension

- Approximately 127 MPa of Stress
- PETG Failed in Simulation
- Upgraded to Nylon-CF (Onyx)



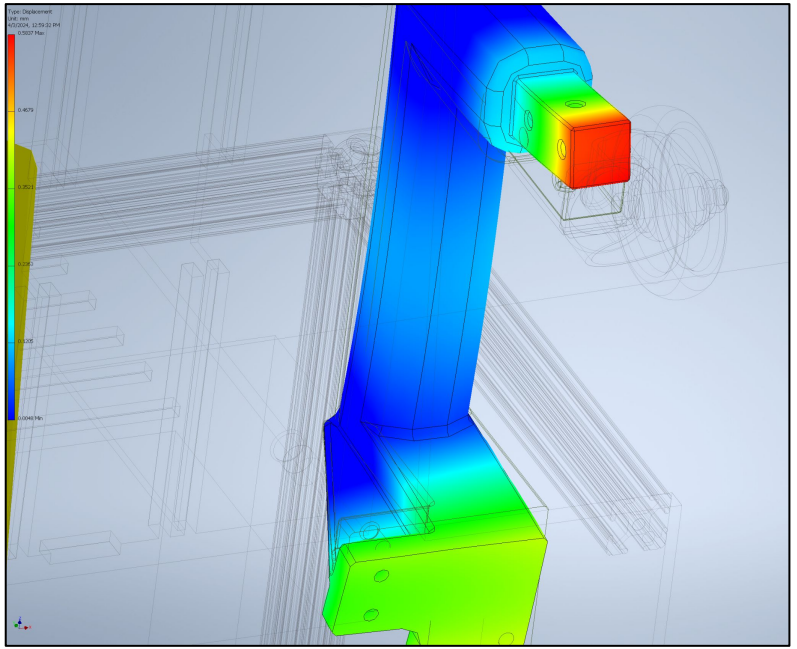


Dynamic Simulation Analysis: Suspension



PETG

Displacement: 14mm

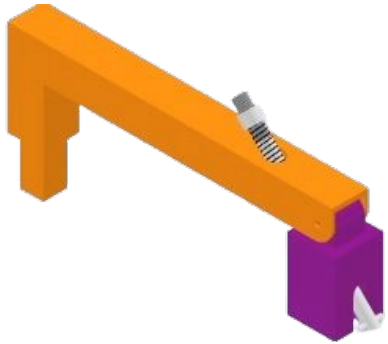


CF Reinforced Nylon

Displacement: 0.4mm

Suspension: Design Iterations

Design 1



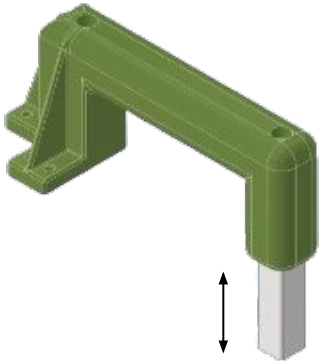
- Lateral Deflection

Design 2



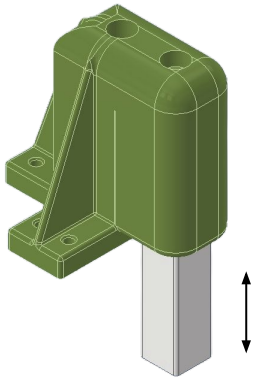
- Vibration

Design 3



- Wheel Deflection

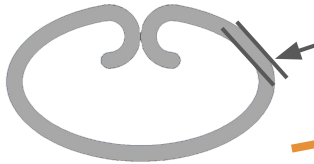
Final Design

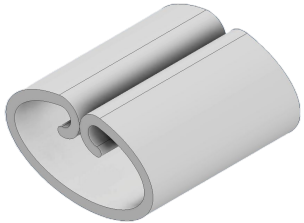


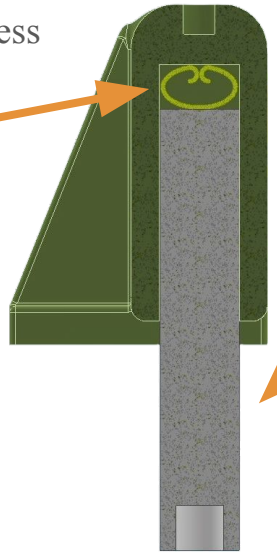
- Acceptable Deflection/ Vibration

Suspension: Design

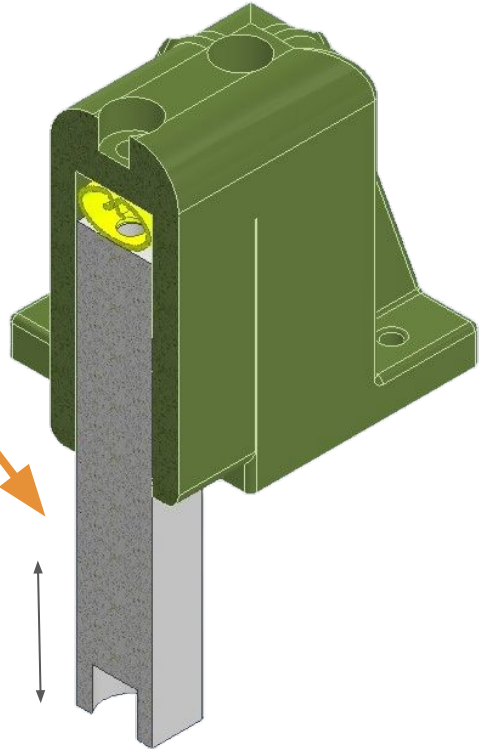
Spring Design:







Slide



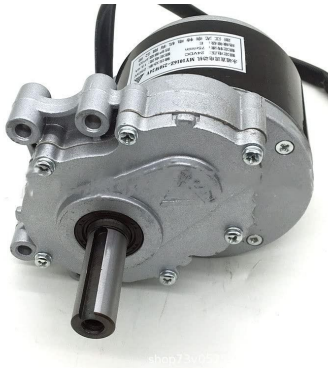
- Spring wall thickness determines spring rate

Motor Choice

- Unitemotor MY1016Z
  - 120 RPM Powered Wheelchair DC Motors
  - 12” Wheel Hardware-Governs Rock-E at Theoretical **4.28 mph**
  - Pair Provides **Up to 80 N.m** at **21.36 A Peak**







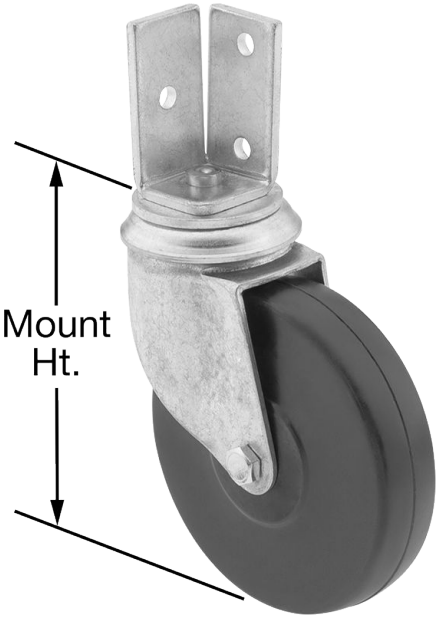


Wheels and Casters

- 12” Polyurethane Wheelchair Tires on Aluminum Wheels

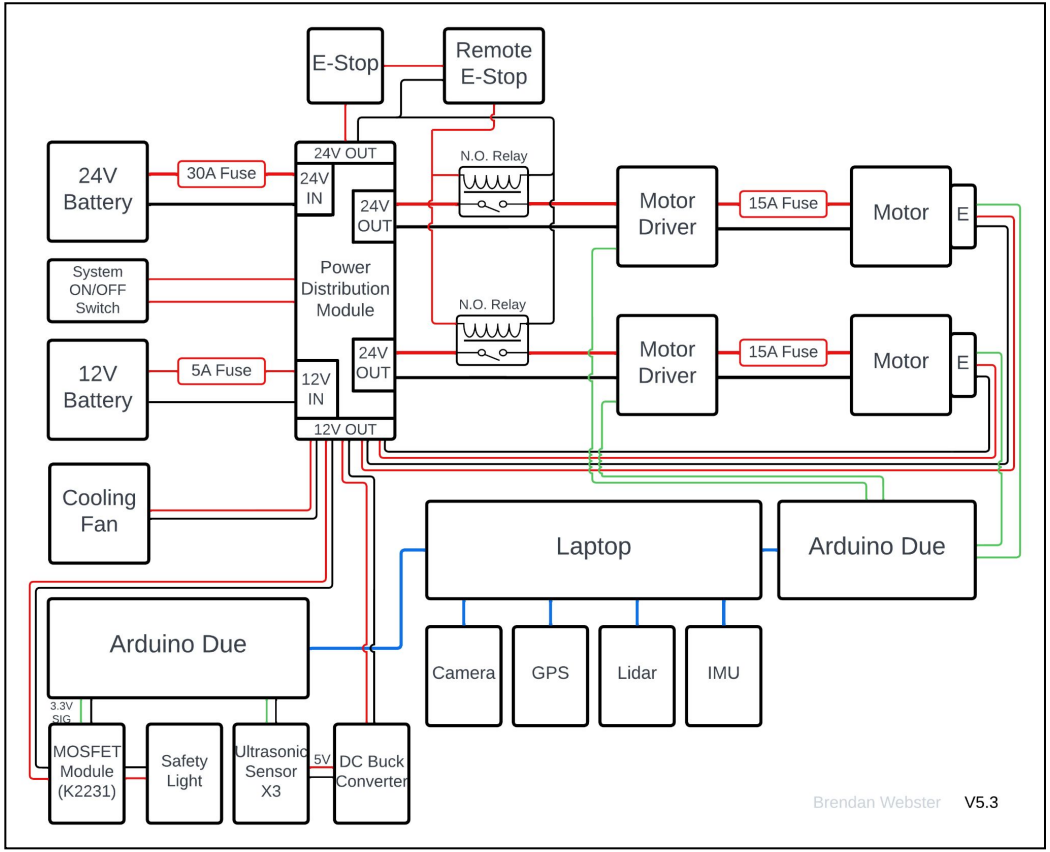


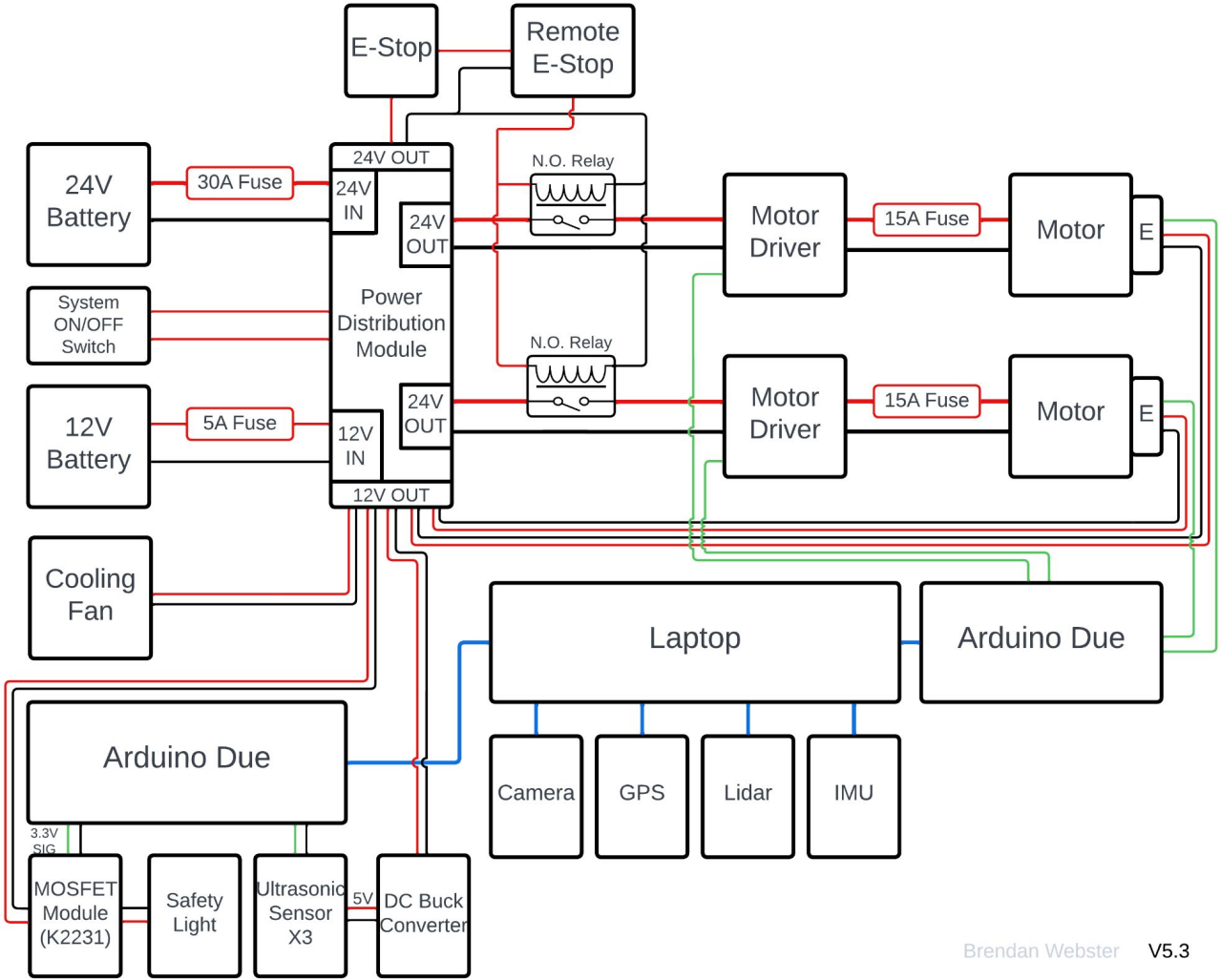
- 3” Hard Rubber Bracket-Mount Swivel Casters



Wiring Diagram

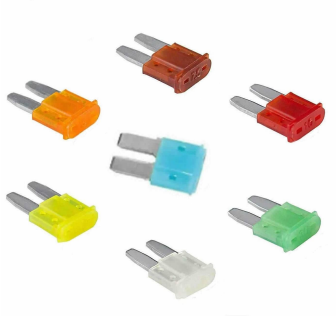
	Positive
	Negative, GND
	Data/Signal
	USB




E-Stop
Remote E-Stop
24V Battery
30A Fuse
24V OUT
24V IN
24V OUT
N.O. Relay
Motor Driver
15A Fuse
Motor
E
System ON/OFF Switch
Power Distribution Module
N.O. Relay
Motor Driver
15A Fuse
Motor
E
12V Battery
5A Fuse
12V IN
24V OUT
12V OUT
Cooling Fan
Laptop
Arduino Due
Arduino Due
Camera
GPS
Lidar
IMU
3.3V
SIG
MOSFET Module (K2231)
Safety Light
Ultrasonic Sensor X3
5V
DC Buck Converter

Safety

Fuses



Mechanical E-Stop



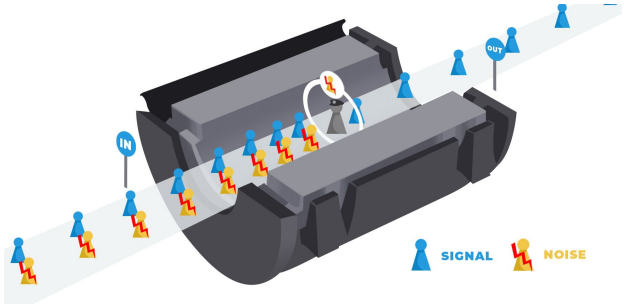
Remote E-Stop



Safety Light



Ferrite Cores



Relays



Batteries





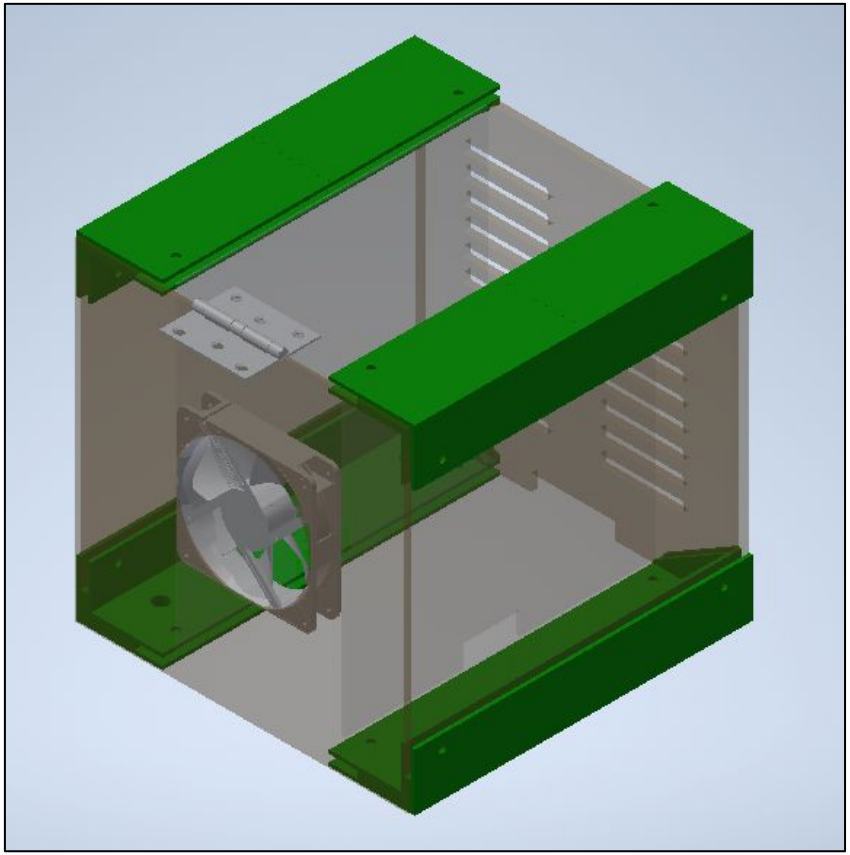
12V Lithium Battery

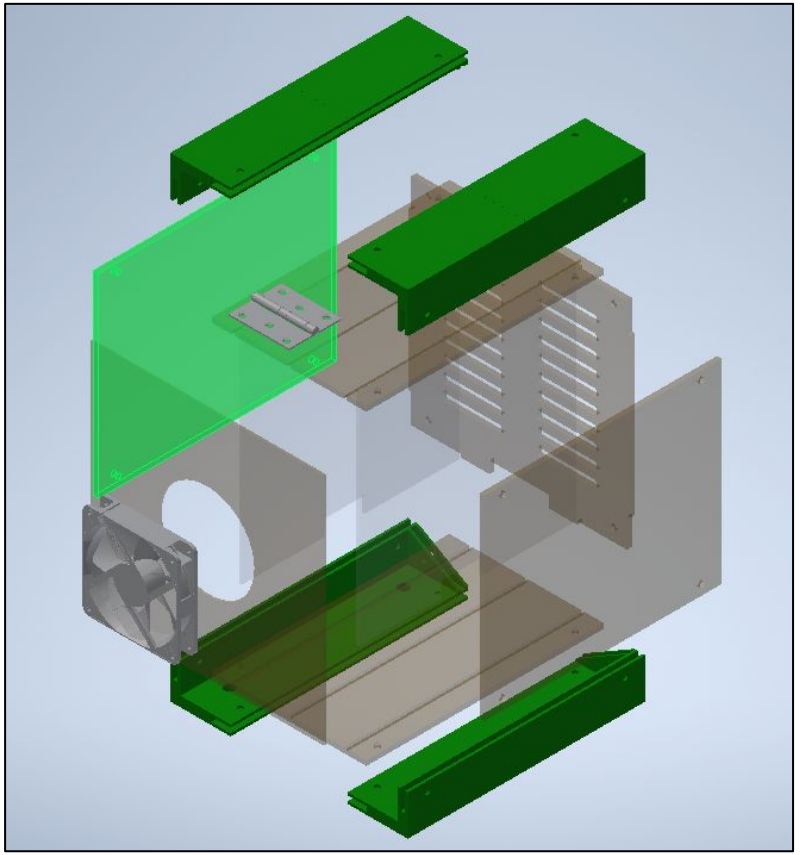
- BMS Circuit Protection
- Battery Level Indicator

24V Lithium Battery

- High Efficiency
- Over 24V at Full Charge

Electronics Box (E-Box)

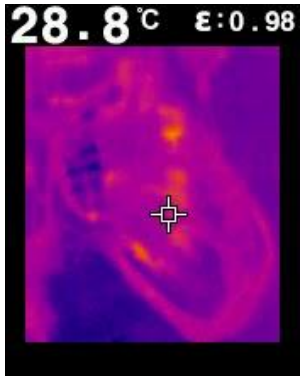


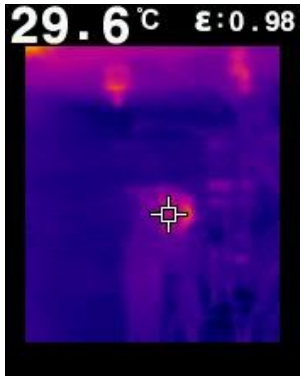


Electronics Box (E-Box)

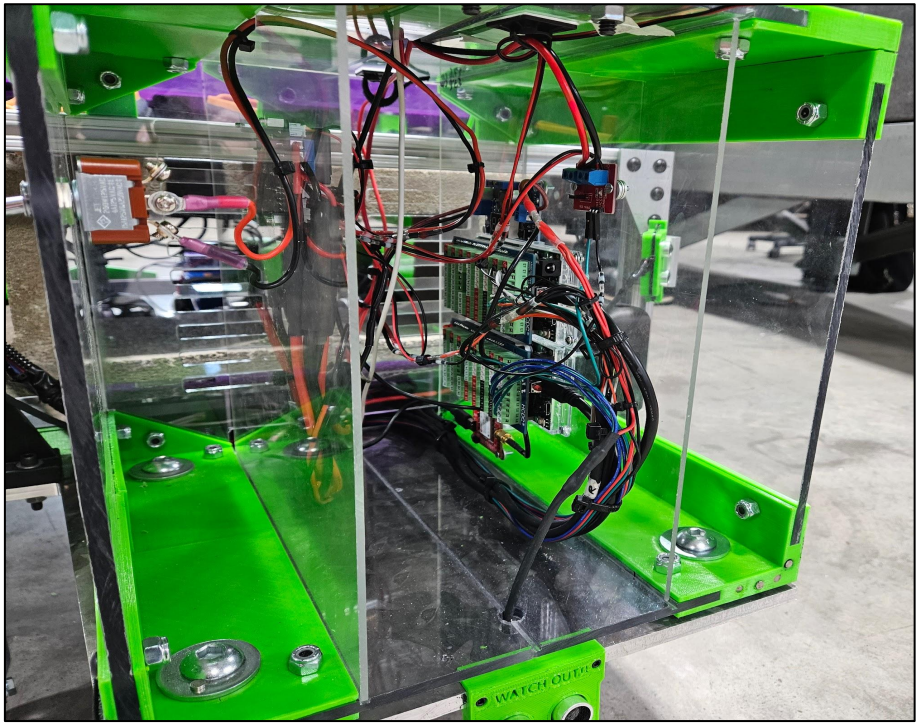
Contains:

- Power Distribution Module
- Arduino Due (x2)
- GPS
- Buck Converter
- MOSFET Circuit





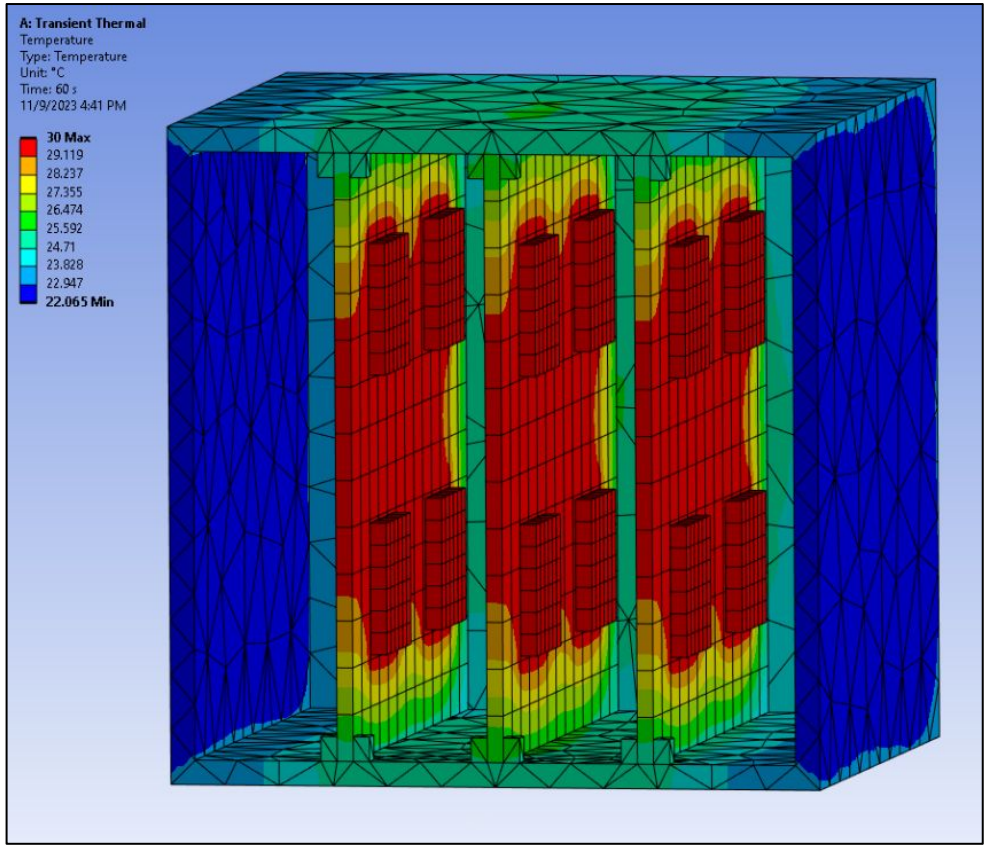
Rear View



Heat and Airflow Simulations

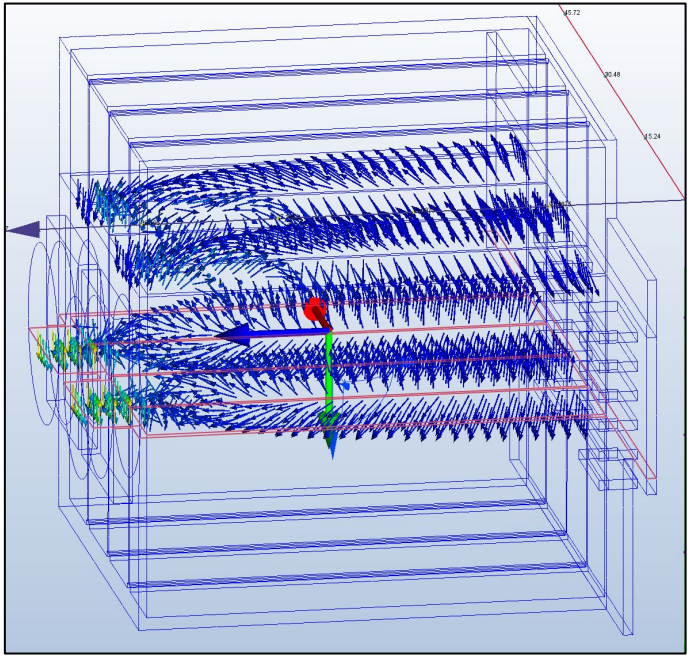
Simulation Boundary Conditions

- Ambient Temperature: **22 °C**
- Operating Temperature: **30 °C**
- Max Temperature Indicated: **30 °C**
- View with Fan Door Removed



Heat and Airflow Simulations

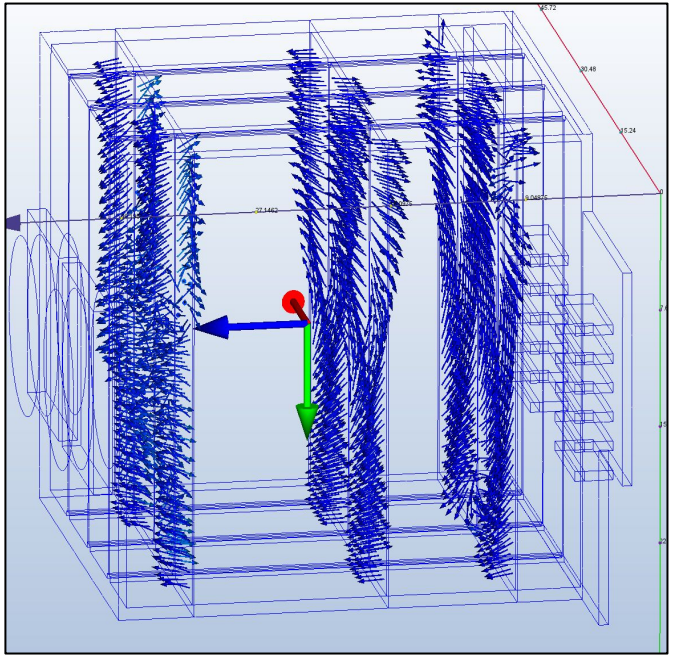
Z-Axis Air Flow



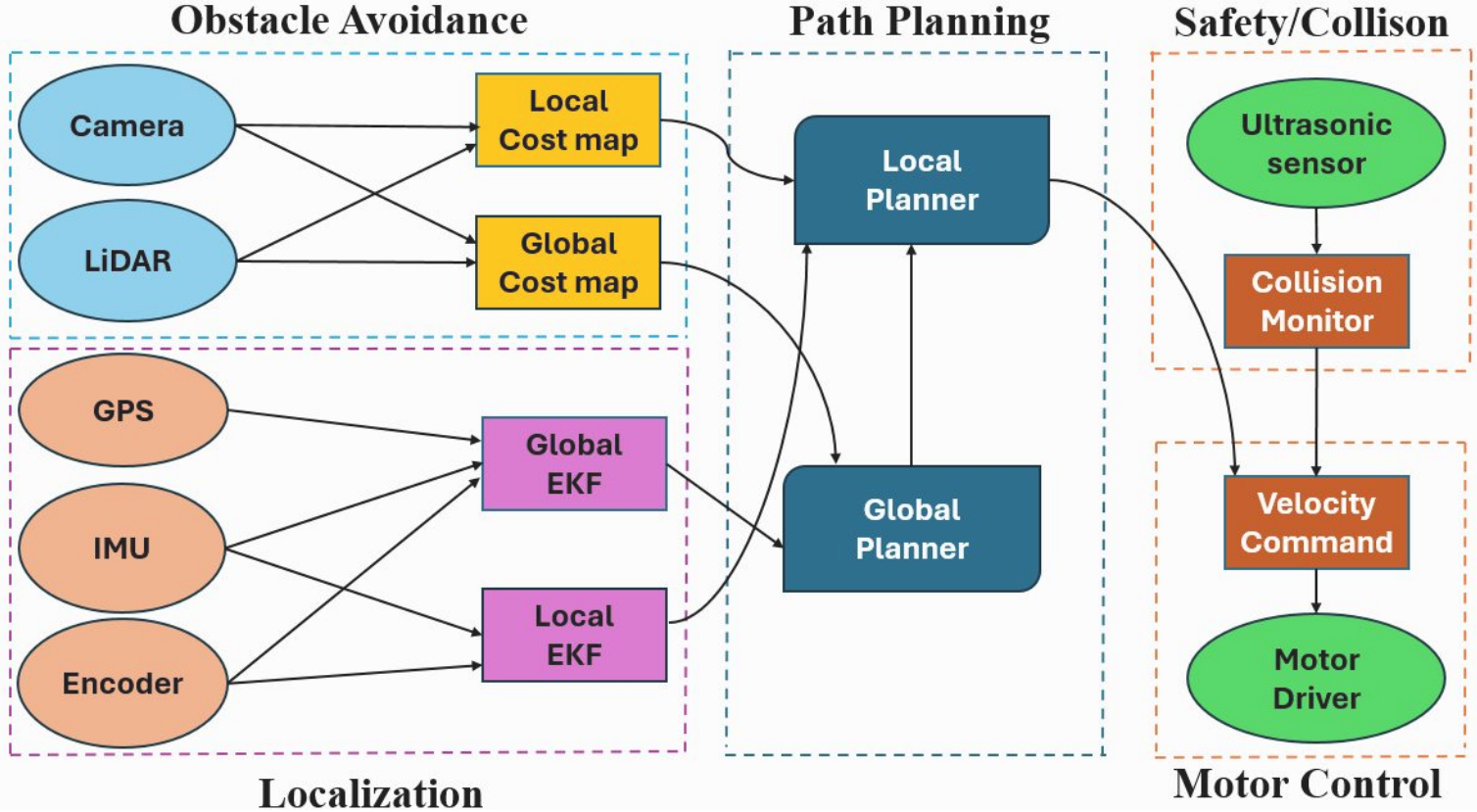
Boundary Conditions:

- 1 mph Inlet
- 1000 rpm Fans
- Side View

X-Axis Air Flow



Final Navigation Prototype

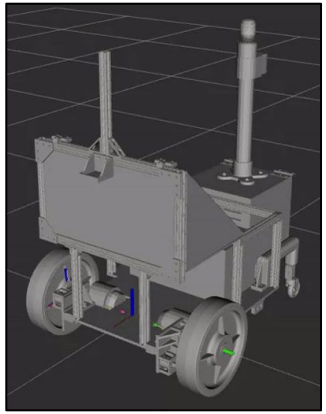


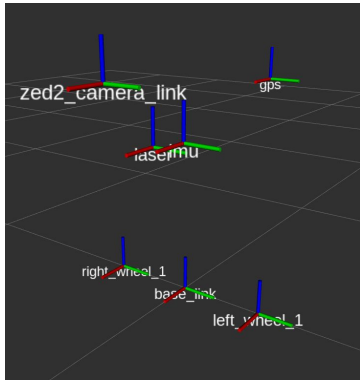
Unified Robotics Description Format (URDF)

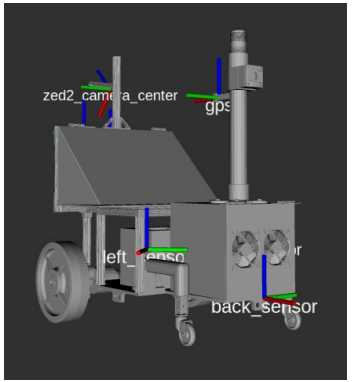
- Robot Model for Simulations
- Used Original CAD Model
- Fusion 360 Export

Transforms and Frames

- URDF Stores Frame Info
- Location and Orientation of Sensors
- Provides Context of Sensor Data



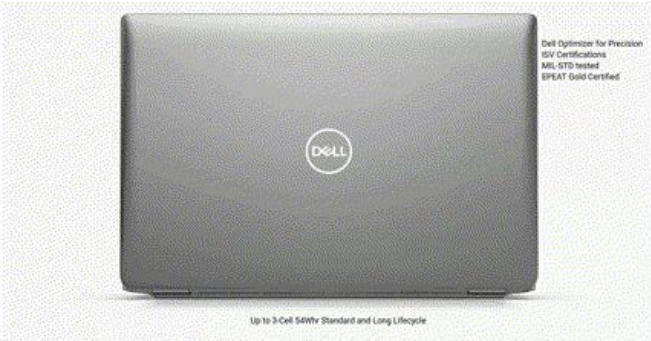




Central Processor

Dell Precision 3480

- Powerful Computer
- Capable of Handling Complex Algorithms
- Reliable Performance for Demanding Tasks

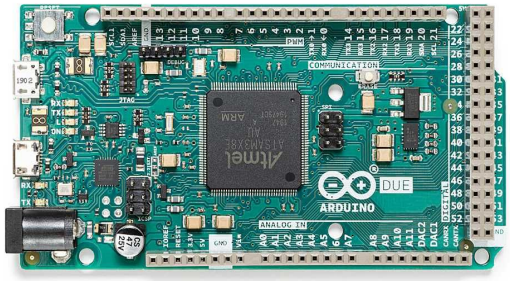


ROS2 - Humble Hawksbill

- Enhanced Real-time Capabilities
- Robust Middleware Software
- Simple to Add and Change Code



Microcontrollers



Purpose

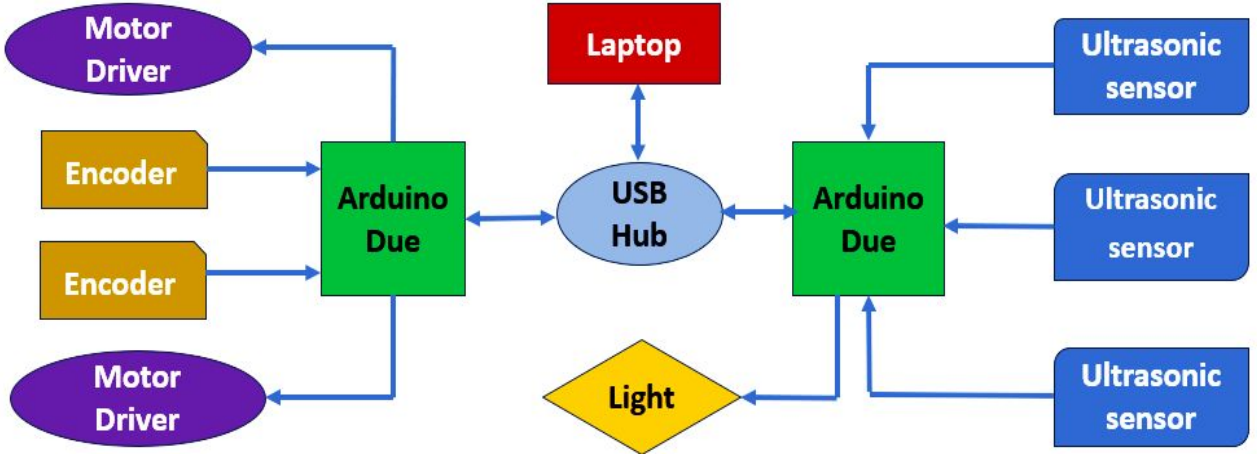
- Less Load, PWM
- Encoder, Drivers
- Sensor, Light

Arduino Due

- ROS2
- 96 KB RAM

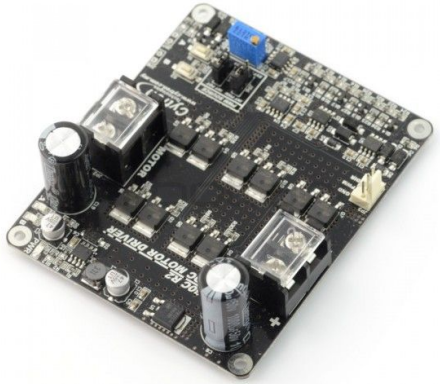
Method

- Micro-ROS Agent



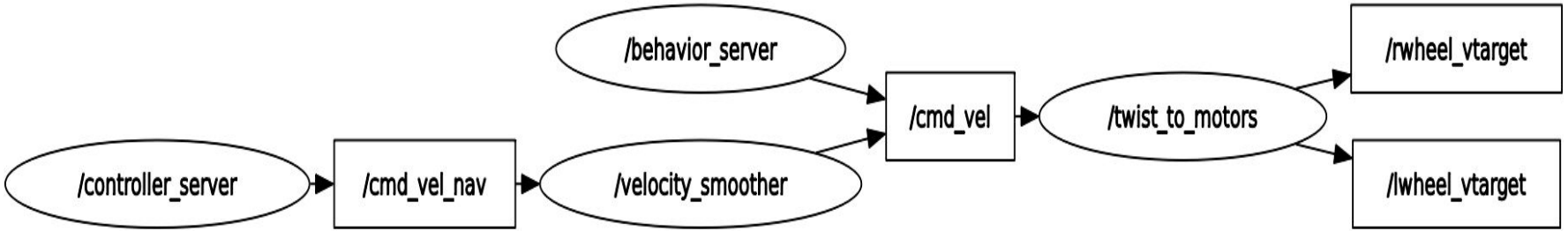
Motor Controls

Cytron MD30C



- 5-30 VDC, 30/80 Amp
- Compatible with Arduino

Velocity Command Data Flow:



Motor Controls

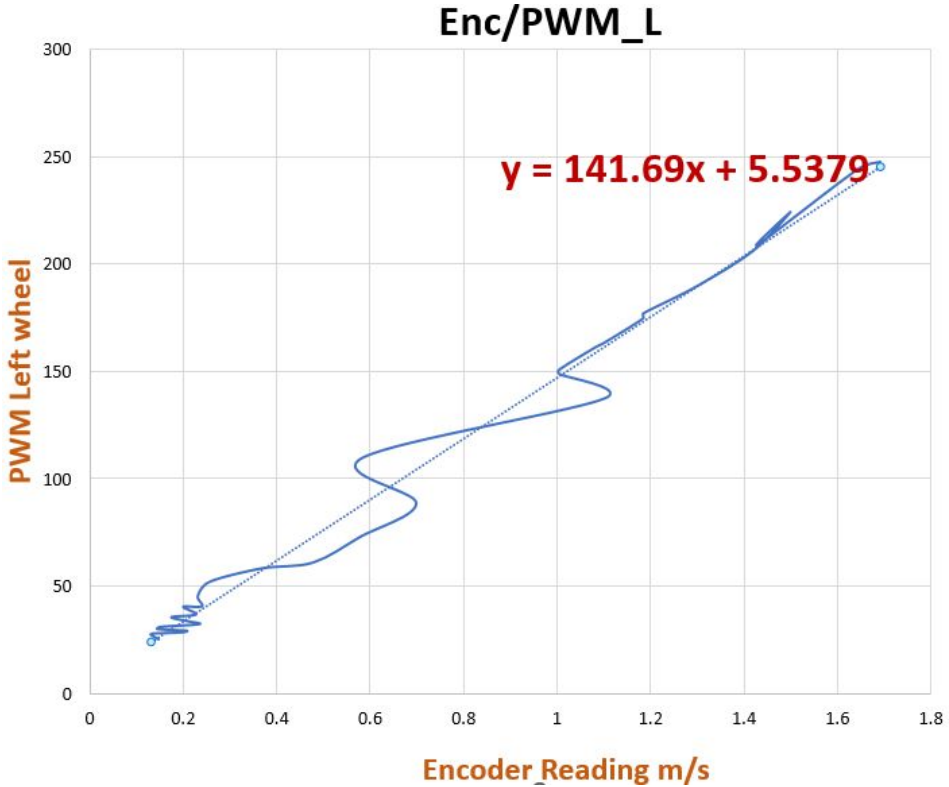
PWM-Linear Constant

Converting

- Velocity Command m/s
- PWM for Motor

Precision

- Encoder Feedback
- Robot Dimension
- Speed Threshold



Remote Control

Purpose

- Joystick Inputs
- Converted to Velocity Commands
- Enable and Turbo Buttons
- Convenient for Moving Robot
- Motor Testing

PlayStation 4 Controller

- Bluetooth Compatible

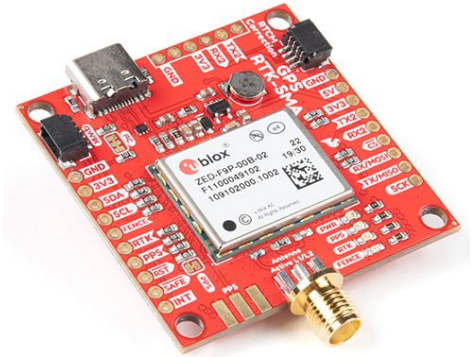




GPS

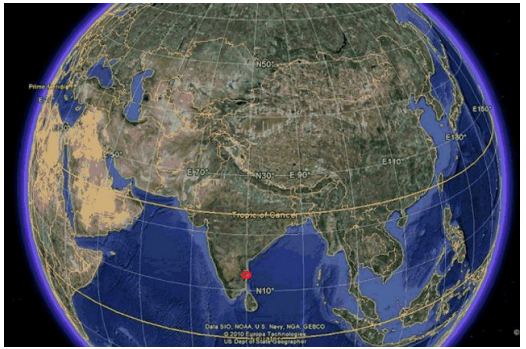
Purpose

- Receiver for GPS Coordinates
- Necessary for Waypoint Navigation
- Fed into Extended Kalman Filter



SparkFun ZED-F9P

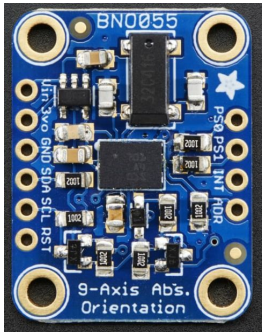
- Accuracy within 2 Meters
- Easily Integrable
- Plentiful Documentation



IMU

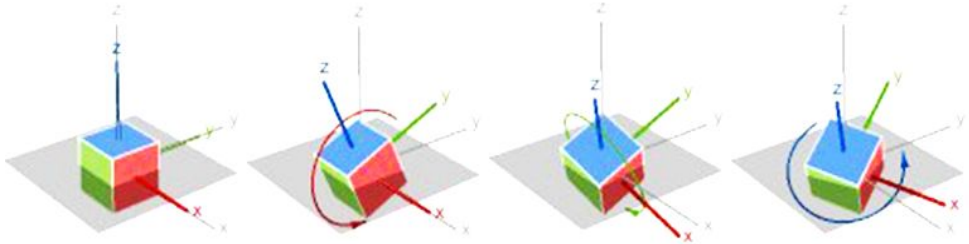
Purpose

- Measures Inertial Values of Vehicle
- Global Heading of Robot
- Fed into Extended Kalman Filter



Adafruit BNO055

- Reasonable Cost
- One Degree of Precision



Encoders

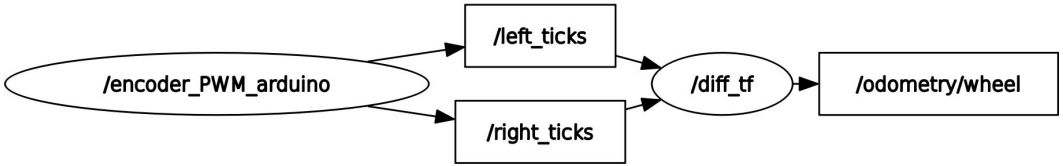
Purpose

- Measures Speed of Wheels
- Converted to Odometry ROS2 Message
- Necessary to be Implemented into EKF



Taiss 360 P/R Rotary Encoder

- Accurate for Price
- Integrable with Motors

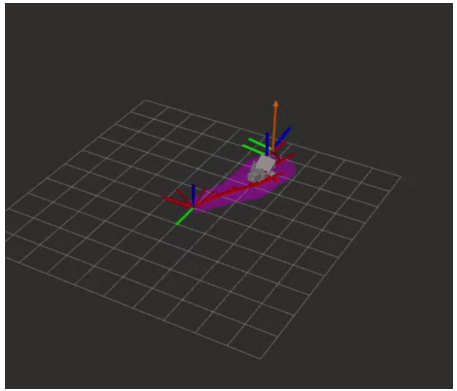


Localization

Extended Kalman Filter (EKF)

- Fuses Data from Multiple Sources
- Local (IMU and Encoders)
- Global (IMU, Encoders, GPS)

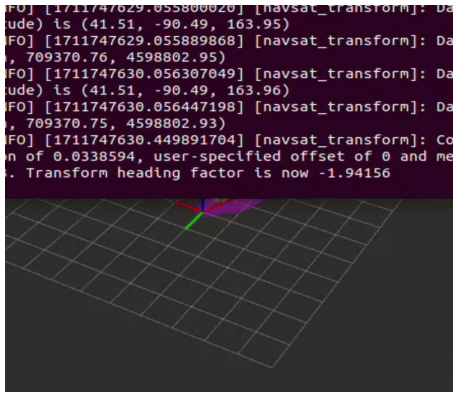
Local



Waypoint Navigation

- Uses Output of Global EKF
- Coordinates and Orientation
- Multiple Points

Global



LiDAR

Purpose

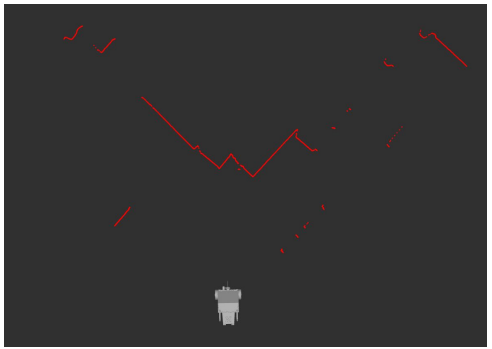
- Obstacle Detection and Avoidance

RPlidar S3



- Outdoor and Indoor Capabilities
- Compatible with ROS 2
- Field of View between 120 to 180 Degrees

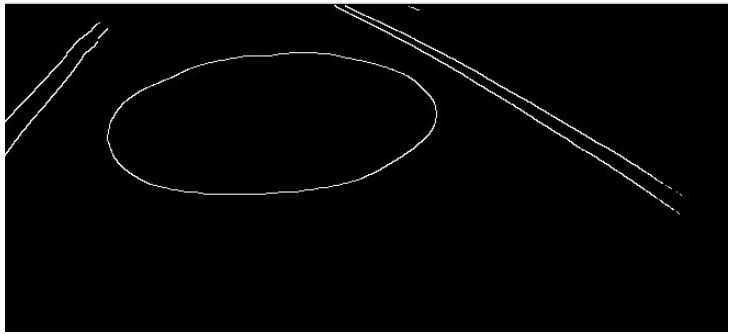




Camera

Purpose

- Line and Pothole Detection



ZED 2i

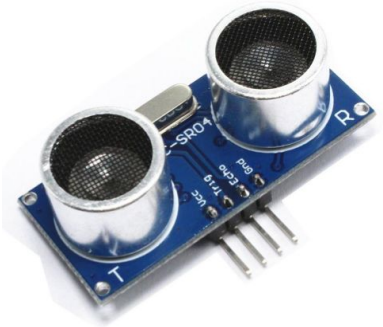
- Compatible with ROS 2
- Performance in Different Light Condition
- RGB Color Ratio
- Real-time Processing Capability



Ultrasonic Sensors

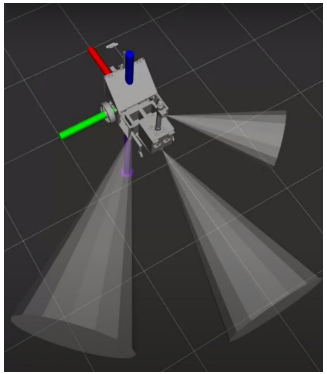
Purpose

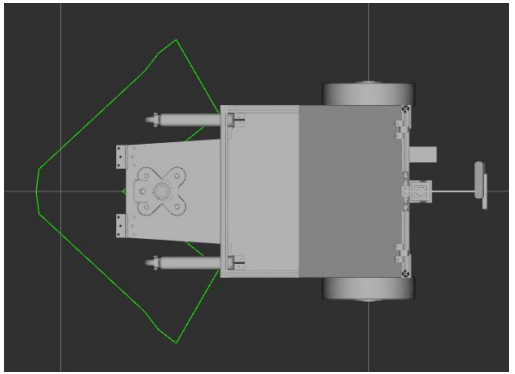
- Proximity Sensors
- Collision Monitoring
- Cover LiDAR's Blind Spots



HC-SR04 Sensor

- Inexpensive
- Readily Available
- Available Libraries and Documentation



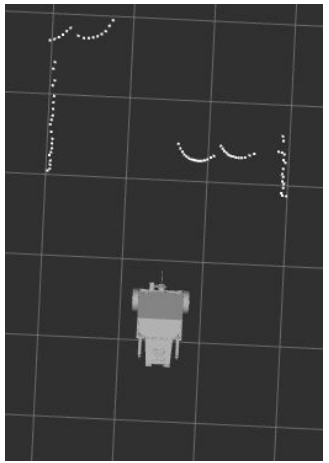


Obstacle Avoidance

Integration of LiDAR, Camera and Ultrasonic Sensors

- Camera Outputs 3D Point Cloud Data
- Converted to 2D Scan for Alignment
- Simplifies Integration with LiDAR Data
- Ensures They Work together Smoothly
- Ultrasonic Sensors Serve as a Fail-safe Mechanism
- Boosts Perception System Abilities





Obstacle Avoidance

SLAM

- Localization and Mapping
- Creates a Map of All Obstacles

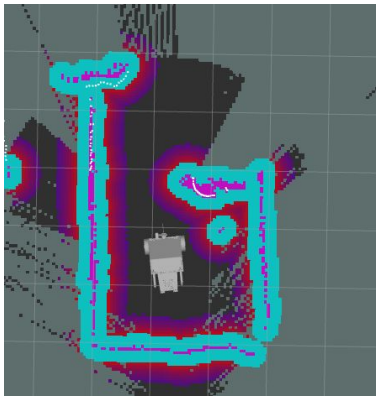
Nav2 Costmaps

- Adds Zones of Cost around Obstacle Map
- Local and Global Costmaps Created









Path Planning

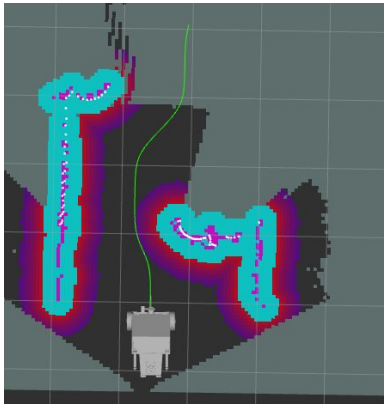
Global Planner

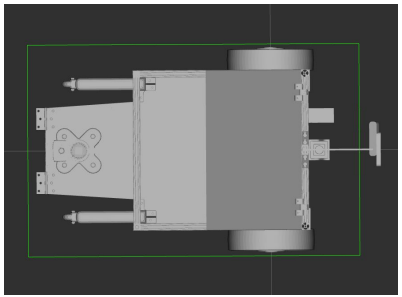
- Smac Planner
- Long-term Waypoint Navigation
- Saves all Cost Data for Map

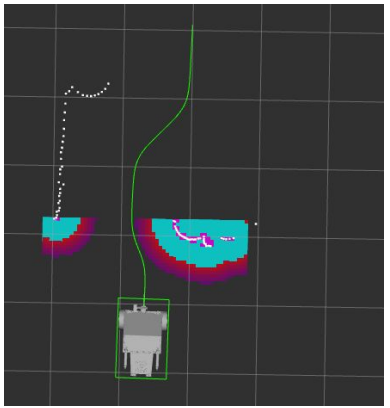
Local Planner

- DWB Controller
- Short-term Obstacle Avoidance
- Provides Velocity Commands

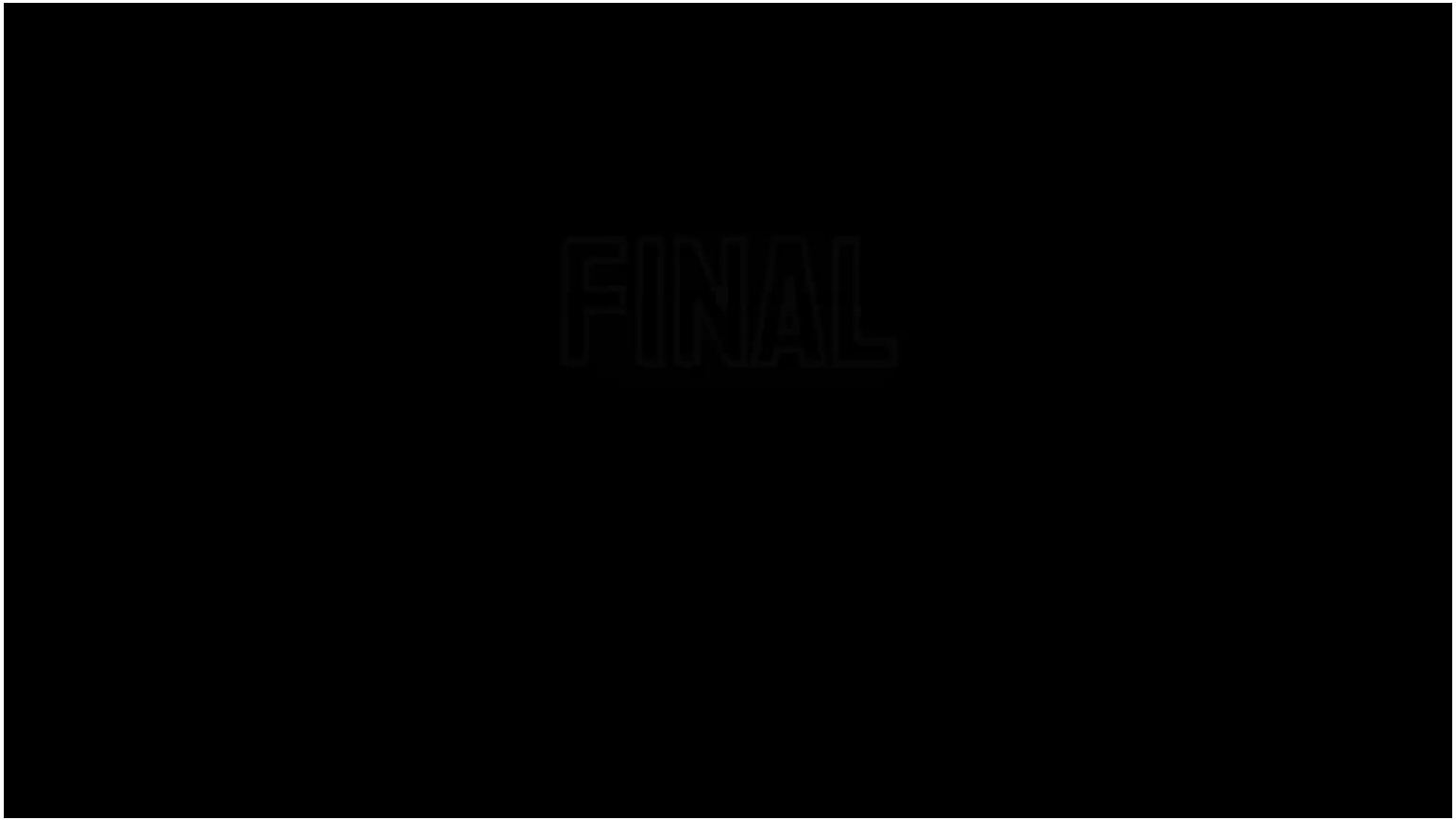








Final Product



Total Expenses:

Purchase Total:

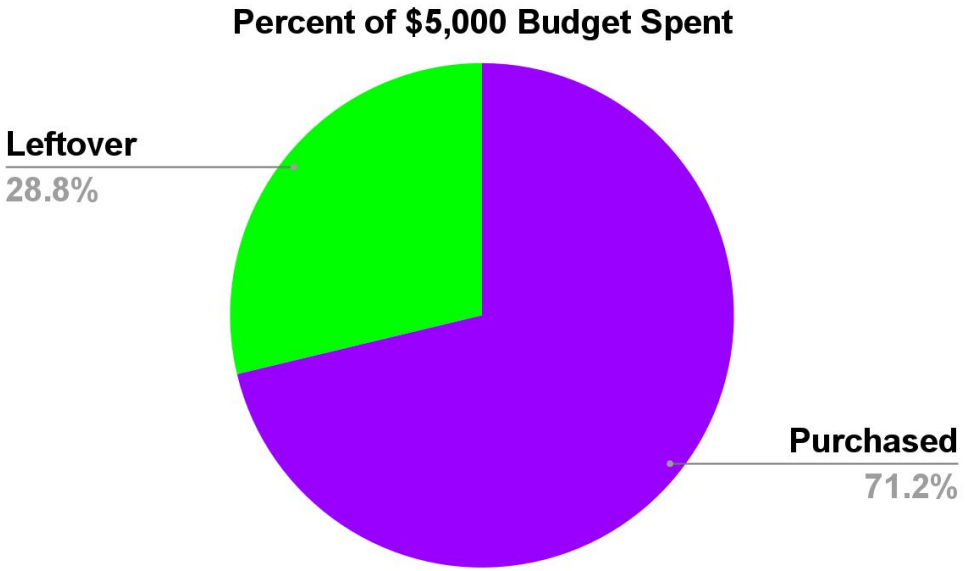
$3,562.48

No Purchase Total:

$3,143.71

Gross Total:

$6,706.19



SET Faculty/Staff:

- Dr. Md Shahin Alam
- Charity Chepchirchir
- Aaliyah Elfridge
- Dr. Il-Seop Shin
- Ashley James
- Rafael Obregon
- Pam Young
- Dr. Khaled Zbeeb
- University Faculty/Staff

THANK YOU!

Sponsors:

LE CLAIRE MANUFACTURING CO

WESTERN ILLINOIS UNIVERSITY 1899

CADDIS

Questions?

